facilities, in “a far away place” whose location, though almost certainly well known to Western authorities by the late 1950s, was forbidden to be revealed by Soviet citizens until 1990.[1] The facility was known simply as *Obyekt* (“the Installation”); Khariton became its director, and Zel’dovich the leader of one of its key bomb design teams. Under Beria’s authority, Kurchatov set up several teams of physicists to pursue, in parallel and completely independently, each aspect of the bomb project; redundancy brings security. The teams at the Installation fed design problems to the other teams, including a small one led by Lev Landau at the Institute of Physical Problems in Moscow.

While this massive effort was rolling inexorably forward, Soviet spies were acquiring, through Klaus Fuchs (a British physicist who had worked on the American bomb project), the design of the Americans’ plutonium-based bomb. It differed somewhat from the design that Zel’dovich and his colleagues had produced, so Kurchatov, Khariton, and company faced a tough decision: They were under excruciating pressure from Stalin and Beria for results, and they feared the consequences of an unsuccessful bomb test in an era when failure often meant execution; they knew that the American design had worked at Alamogordo and Nagasaki, but they could not be completely sure of their own design; and they possessed enough plutonium for only one bomb. The decision was clear but painful: They put their own design on hold[2] and converted their crash program over to the American design.

At last, on 29 August 1949—after four years of crash effort, untold misery, untold deaths of slave-labor zeks, and the beginning of an accumulation of waste from nuclear reactors near Cheliabinsk that would explode ten years later, contaminating hundreds of square miles of countryside—the crash program reached fruition. The first Soviet atomic bomb was exploded near Semipalatinsk in Soviet Asia, in a test witnessed by the Supreme Command of the Soviet army and government leaders.

On 3 September 1949 an American WB-29 weather reconnaissance plane, on a routine flight from Japan to Alaska, discovered products of nuclear fission from the Soviet test. The data were given to a committee of experts, including Oppenheimer, for evaluation. The verdict was unequivocal. The Russians had tested an atomic bomb!